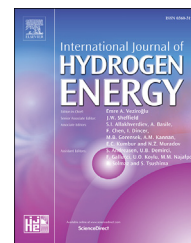


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Chemical kinetic modeling of ammonia oxidation with improved reaction mechanism for ammonia/air and ammonia/hydrogen/air combustion

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ABSTRACT

To achieve comprehensive prediction of ammonia combustion in terms of flame speed and ignition delay time, an improved mechanism of ammonia oxidation was proposed in this work. The present model (UT-LCS) was based on a previous work [Song et al., 2016] and improved by relevant elementary reactions including NH_2 , HNO , and N_2H_2 . The model clearly explained reported values of laminar flame speed and ignition delay time in wide ranges of equivalence ratio and pressure. This suggests that NH_2 , HNO , and N_2H_2 reactivities play a key role to improve the reaction mechanism of ammonia oxidation in the present model. The model was also applied to demonstrate $\text{NH}_3/\text{H}_2/\text{air}$ combustion. The present model also appropriately predicted the laminar flame speed of $\text{NH}_3/\text{H}_2/\text{air}$ combustion as a function of equivalence ratio. Using the model, we discussed the reduction of NO concentration downstream and H_2 formation via NH_3 decomposition in NH_3/H_2 fuel-rich combustion. The results provide suggestions for effective combustion of NH_3 for future applications.

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Introduction

Transition to a future society that is not dependent on fossil fuels is becoming an extremely important goal for reducing CO_2 emissions. Investigations of alternatives to carbon-based liquid fuels and hydrogen carriers are thus crucial

challenges. Ammonia, which is a carbon-free fuel, is a promising green energy carrier/storage medium because it has high energy density and can be easily liquefied [1–5]. In parallel, ammonia combustion has been examined for applications in gas turbines and gas engines in terms of fuel flexibility [6–13]. However, effective use of ammonia fuel remains challenging

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because the minimum required ignition energy is high and the burning velocity is very slow in comparison with conventional hydrocarbon fuels [3,14,15]. Recently, despite these difficulties, an essential advance in ammonia combustion with a micro gas turbine was reported [14]. Stable operation of ammonia–air power generation using a 50-kW-class micro gas turbine system was demonstrated successfully using a diffusion-flame-type combustor [14]. Therefore, application of ammonia fuel in a gas turbine is coming closer to reality. For practical use, however, further improvements in output power and efficiency, and reducing NO_x emissions are indispensable.

To provide solutions to effective ammonia combustion in gas turbines and gas engines, tremendous efforts through experimental observations and kinetic modeling have been made to understand properties such as combustion temperature, flame seed, ammonia and NO_x concentrations in pure ammonia combustion [16–19], ammonia/methane mixture combustion [7,20–23], and ammonia/hydrogen mixture combustion [3,24–33]. Detailed description of appropriate combustion kinetic mechanisms is important for understanding the reaction paths of ammonia decomposition and exhaust-gas formation.

Ammonia combustion models have been proposed and improved by several research groups [15,18,21,34–45]. In early studies, specific ammonia oxidation models were proposed to describe ammonia flame burning [42] and flat laminar premixed flame in ammonia combustion [43]. Later, Konnov and De Ruyck proposed a detailed mechanism to discuss ammonia oxidation decomposition, ignition, and flame structure [34,37,44]. The model of Konnov involved a full H/N/O mechanism (129 species, 1231 reactions [44]) and was tested widely and corrected in subsequent studies on ammonia combustion [3,19,27,28,36,46]. Other models were proposed by Miller-Bowman [15], GRI [38], and San Diego [47] for natural gas burning, taking into account nitrogen species such as ammonia and nitrogen monoxide. Those models were also examined for ammonia oxidation [19,32,33]. The predicted and measured flame speed were in good agreement [19,32], but some disagreements were observed for stoichiometric conditions with small ammonia concentrations and fuel-lean conditions [32]. To mitigate the discrepancies between predictions and experiments, additional discussions are ongoing [21,26,27,32,35,36,39,40,46]. To optimize operating conditions of ammonia combustion, more accurate predictions are necessary in wider ranges of temperature, pressure, and stoichiometry. To date, accurate predictions of flame speed, flame structure, and NO_x concentration have not been achieved under a variety of pressurized and stoichiometric, i.e., from fuel-lean to fuel-rich conditions, but this information is indispensable. Further improvements of the models are required for practical applications to gas turbines and gas engines with high efficiency and low NO_x emission concentrations.

Here, we describe the current status of predictions of ammonia combustion properties. In Fig. S1 of the supplementary material, laminar flame speed is shown as a function of fuel–air equivalence ratio using previous experimental [19,48–51] and numerical calculation [18,21,37,40,52,53] results. The fuel–air equivalence ratio is defined by overall reaction, $\text{NH}_3 + 3/4\text{O}_2 \rightarrow 1/2\text{N}_2 + 3/2\text{H}_2\text{O}$, in this study. The calculated values of laminar flame speed were higher than

experimental values. Although recently calculated values, such as those reported by Song et al. [40], were improved in comparison with the early work by Konnov [37], the deviation between experiments and calculations remains.

In Fig. S2 of the supplementary material, we compare the shock tube ignition delay times of experimental values (Mathieu and Petersen [18]) and calculated values (Song et al. [40], Tian et al. [21], Dagaut et al. [52], Klippenstein et al. [53], and Mathieu and Petersen [18]). Models proposed by Song et al. [40], Tian et al. [21] and Klippenstein et al. [53] resulted in similar results, i.e., calculated values were smaller than the experimental values, as shown in Fig. S2. However, good agreement between experiments and calculations was achieved by Dagaut et al. [52], as previously indicated by Klippenstein et al. [53], and Mathieu and Petersen [18]. Thus, these models explained the ignition delay time well, but did not reproduce experimental results for flame speed, as shown in Fig. S1. Thus, all previous works have overestimated ammonia combustion in terms of flame speed. Mathieu and Petersen [18] refined the model of Dagaut et al. [52]. They added relevant elementary reactions and changed reaction rate constants to improve the model Mathieu and Petersen [18] in Fig. S2. This model explained the ignition delay time very well, but also overestimated the flame speed of ammonia combustion, as shown in Fig. S1. In the present study, therefore, we propose an ammonia combustion model to explain both flame speed and ignition delay time satisfactorily.

In the present study, we improve a chemical kinetic model of ammonia combustion through a preliminary sensitivity analysis based on a previous model [40]. We propose a modified model of ammonia combustion, which provides more accurate predictions of flame speed, ignition delay, and concentrations of ammonia and NO_x in an exhaust gas in a wide range of operating conditions, including pressure and stoichiometry (i.e., equivalence ratio of ammonia/oxygen and ammonia/hydrogen/oxygen). Using the modified model, we discuss conditions for effective ammonia combustion to achieve low NO_x emission, which can provide strategies for developing future applications of NH₃ combustors.

Computational details

In this study, the improved reaction model of ammonia oxidation (UT-LCS model) was made using a base model developed by Song et al. (32 species, 204 reactions) [40], which was, in turn, a modified model based on the model of Klippenstein (31 species, 202 reactions) [53]. The proposed reaction model in this study was examined using CHEMKIN PRO software package [54].

The details of improvements of an ammonia combustion model are as follows. As Song and co-researchers have suggested, reactions of amino radical are important in ammonia combustion. We thus developed the model, taking into this point. First, we conducted a sensitivity analysis of elementary reactions for laminar flame seed using the model of Song et al. (see Table S1 in the supplementary material) with equivalence ratio $\phi = 1.1$ at atmospheric pressure. The definition of a sensitivity coefficient for laminar flame speed of reaction j , S_j , is as follows.

$$S_j = \frac{A_j}{u} \frac{\partial u}{\partial A_j} = \frac{\partial \ln u}{\partial \ln A_j}, \quad (1)$$

where A_j is the pre-exponential factor of the reaction j and u is the liner velocity of the unreacted gas in the flame fixed coordinate (i.e., laminar flame speed). We conducted sensitivity analysis for laminar flame speed and listed elementary reactions which had relatively large values of sensitivity coefficients (see Table S1 for details). For these reactions, we performed a literature survey and selected rate constants which best reproduced the reported values of laminar flame velocity (Fig. S1). Therefore, we changed rate constants based on reported values, but did not tune relevant rate constants. The changed values are listed in Table 1 (R1–R7).

Next, we also considered pressure dependence for relevant elementary reactions. Dean and Bozzelli evaluated the rate constants of unimolecular decompositions for NH_3 , N_2H_2 , and H_2NN using Quantum Rice-Ramsperger-Kaseel (QRRK) analysis with the master equation (ME) analysis [55]. The pressure dependence of the rate constants was expressed by PLOG functions implemented in the CHEMKIN PRO software package. The reactions are summarized in Table 1 (R8–R10).

In addition, since some rate constants could play important roles in a high temperature region, we added the rate constants in the present model (Table 1 (R11–R23)). Also, we added reactions R24–31 in Table 1. The detailed reaction mechanism (full mechanism) is described in the supplementary material (UT-LCS chem.inp, therm.dat, and tran.dat attached as input files for CHEMKIN PRO in the supplementary material).

Ignition delay time was calculated in the following manner. In experimental work performed by Mathieu and Petersen [18], ignition delay time was measured by observing the time profile of an excited state hydroxyl radical (OH^*), i.e., measuring the onset time of chemiluminescence emission of OH^* from the slope of OH^* time profile in a reflected shock wave to determine the ignition delay time. In the present study, assuming adiabatic and constant volume conditions, we calculated OH^* time profiles considering the excitation process of OH^* , $\text{O} + \text{H} + \text{M} \rightarrow \text{OH}^* + \text{M}$ and $\text{N}_2\text{O} + \text{H} \rightarrow \text{N}_2 + \text{OH}^*$, and the deactivation process of OH^* , $\text{OH}^* + \text{M} \rightarrow \text{OH} + \text{M}$ ($\text{M} = \text{Ar}, \text{H}_2, \text{O}_2, \text{OH}, \text{H}, \text{O}$) according to the model by Mathieu and Petersen [18]. Then we calculated ignition delay time τ_{OH^*} , as well as the experimental manner by Mathieu and Petersen. In addition, to simplify the procedure for calculation of τ_{OH^*} , we evaluated ignition delay time $\tau_{\Delta T}$, defined as the elapsed time taken for the temperature to increase by 50 K from the initial temperature. Because $\tau_{\Delta T}$ corresponded well to τ_{OH^*} , we employed $\tau_{\Delta T}$ as ignition delay time in this study, instead of τ_{OH^*} .

Results and discussion

Model validation

Shock tube ignition delay data obtained by Mathieu and Petersen [18] are compared with simulation results in Fig. 1. Ignition delay times calculated by the present model agree very well with experimental data over wide ranges of equivalence

ratios and pressures. The performance of the present model is very similar with the model of Dagaut et al. [52] and its modified version of Mathieu and Petersen [18] (see Fig. S2 in the supplementary material). As can be seen in Fig. 1, the present model gives better agreement with experimental data than the base model proposed by Song et al. [40]. The improvement of the present model over the base model is caused by the changes of rate constants for reactions 1–23 and inclusion of reactions 24–31 in Table 1. In order to clarify the reason of this improvement, the sensitivity analysis for temperature has been performed. An example is shown in Fig. S3 in the supplementary material. Reactions (1) $\text{NH}_2 + \text{H} = \text{NH} + \text{H}_2$, (7) $\text{NH}_2 + \text{NH} = \text{N}_2\text{H}_2 + \text{H}$, (9) $\text{N}_2\text{H}_2 = \text{NNH} + \text{H}$, and (11) $\text{NH} + \text{H} = \text{N} + \text{H}_2$ in Table 1 are sensitive to temperature.

Although models proposed by Dagaut et al. [52] and Mathieu and Petersen [18] can reproduce experimental shock tube ignition delay times quite well, these models considerably overestimate laminar flame speeds as shown in Fig. S1 in the supplementary material. Other previous models proposed by Klippenstein et al. [53,58], Tian et al. [21] and Song et al. [40] (the base model of the present study) also overestimate laminar flame speed. Agreement with experimental flame speeds is greatly improved in the present model as shown in Fig. 2. Sensitivity analysis on flow rate indicates that this improvement is due to changes of rate constants for reactions of 1, 2, 4, 7, 8, and 20 in Table 1. These include NH_2 and NH reactions which are important for NO production rates as discussed in the next section. As a result, time profile of NO is different between the present model and the base model.

The difference of flame speed between the present model and the base model is partly due to the unimolecular decomposition reaction of N_2H_2 (reaction 9 in Table 1),



Since N_2H_2 is mainly produced by the $\text{NH}_2 + \text{NH}_2 = \text{N}_2\text{H}_2 + \text{H}_2$ reaction (reaction 5), $\text{NH}_2 + \text{NH}_2$ reactions also have considerable effect on NH_3 combustion, especially at rich conditions. The $\text{NH}_2 + \text{NH}_2$ reactions are multi-channel, chemically activation reactions, including following channels:



In principal, branching ratios of these channels are all pressure dependent. Asatryan et al. [62] had performed a comprehensive study on the formation and decomposition of chemically activated N_2H_4 on the basis of quantum chemical calculations (CCSD(T)/CBS level of theory). In their study, they estimated branching ratios of $\text{NH}_2 + \text{NH}_2$ reaction by using QRRK/Master equation analysis. At high temperatures above 1000 K and $p = 1$ atm, the main product channel is $\text{NH}_3 + \text{NH}$. Unfortunately, analytical form of the rate constants for each product channels was not given as a function of pressure. In the present model, pressure independent rate constants proposed by Klippenstein et al. [53,58] based on their theoretical study. Inclusion of pressure dependence of rate constants for

Table 1 – List of reactions modified in or added to the mechanism from Song et al. [40].

Reaction		Rate coefficient (cm, mol, s, cal)			Source
		A	n	E	
1	NH ₂ + H = NH + H ₂	1.00E+06	2.32	799	[36]
2	NH ₂ + NO = NNH + OH	4.29E+10	0.29	−870	[56]
3	NH + NO = N ₂ O + H	5.00E+14	−0.4	0	[36]
4	HNO + H=NO + H ₂	9.68E+11	0.62	356	[57]
5	NH ₂ + NH ₂ = N ₂ H ₂ + H ₂	1.70E+08	1.02	11783	[58]
6	NH ₂ + NH ₂ = H ₂ NN + H ₂	7.20E+04	1.88	8802	[58]
7	NH ₂ + NH = N ₂ H ₂ + H	1.50E+15	−0.5	0	[59]
8	N ₂ H ₃ = N ₂ H ₂ + H				[55] ^a
	0.1 atm	2.30E+43	−9.55	64468	
	1.0 atm	3.60E+47	−10.38	69009	
	10 atm	1.80E+45	−9.39	76102	
9	N ₂ H ₂ = NNH + H				[55] ^a
	0.1 atm	5.60E+36	−7.75	70340	
	1.0 atm	1.80E+40	−8.41	73320	
	10 atm	3.10E+41	−8.42	76102	
10	H ₂ NN = NNH + H				[55] ^a
	0.1 atm	5.90E+32	−6.99	51791	
	1.0 atm	9.60E+35	−7.57	54841	
	10 atm	5.00E+36	−7.43	57295	
	DUPLICATE				
	0.1 atm	7.20E+28	−7.77	50758	
	1.0 atm	3.20E+31	−6.22	52318	
	10 atm	5.10E+33	−6.52	54215	
11	NH + H = N + H ₂	1.00E+14	0	0	[15,18]
12	HNO + O = NO + H	1.00E+13	0	0	[18,60]
13	HNO + HNO = N ₂ O + H ₂ O	3.95E+12	0	5000	[18,60]
14	NO ₂ + NO ₂ = NO + NO + O ₂	2.00E+12	0	2685	[18,60]
15	HONO + OH = NO ₂ + H ₂ O	1.30E+10	1	135	[18,60]
16	HNO ₂ + H = NO ₂ + H ₂	2.40E+08	1.5	5087	[18,60]
17	HNO ₂ + O = NO ₂ + OH	1.70E+08	1.5	3020	[18,60]
18	HNO ₂ + OH = NO ₂ + H ₂ O	1.20E+06	2	−596	[18,60]
20	N ₂ O(+M) = N ₂ +O(+M)	9.90E+10	0	57960	[18,61]
	Low pressure limit	6.72E+14	0	57500	
	Third body efficiency	N ₂ /1.7/O ₂ /1.4/H ₂ O/12/			
21	N ₂ O + H = N ₂ + OH	3.31E+10	0	5090	[18,61]
	DUPLICATE	7.83E+14	0	19390	
22	N ₂ O + OH = N ₂ + HO ₂	2.00E+12	0	40000	[18,36]
23	HNO + NO = N ₂ O + OH	2.00E+12	0	26000	[18,60]
24	H ₂ NO + M = H ₂ +NO + M	3.83E+27	−4.29	60300	[18,60]
25	NO + HO ₂ + M = HONO ₂ + M	2.23E+12	−3.5	2200	[18,60]
26	NO ₂ + NO = N ₂ O + O ₂	1.00E+12	0	60000	[18,60]
27	N + NO ₂ = N ₂ O + O	1.80E+12	0	0	[18,60]
28	HNO + N = NO + NH	1.00E+13	0	1990	[18,60]
29	N ₂ O + N = N ₂ + NO	1.00E+13	0	19870	[18,60]
30	N ₂ + M = N + N + M	1.89E+18	−0.85	224950	[18,60]
31	N + O + M = NO + M	7.60E+14	−0.1	−1770	[18,60]

^a Pressure dependence of the rate constant is expressed by a PLOG function implemented in CHEMKIN PRO.

these products channels is open for future improvement of the present model.

We performed simulations of atmospheric stagnation flame structures reported by Brackmann et al. [46]. An experimental temperature profile is included in the simulation, and NH, OH and NO profiles are compared. Results are shown in Fig. S4 in the supplementary material. Calculated maximum NH concentration is about 30% higher at $\phi = 0.9$ and 20% lower at $\phi = 1.0$ than experiments. At $\phi = 1.2$, a peak position is much faster and the peak is about 50% higher than the experiment. An OH profile at $\phi = 0.9$ is in good agreement with an experimental profile but 20% higher at $\phi = 1.0$ and 30% lower at $\phi = 1.2$ than experiments. The peak position of OH at $\phi = 1.2$ is faster

than the experiment. Experimental profiles of NO at $\phi = 0.9$ and 1.0 show decay towards a stagnation point ($x = 1.62$ cm), but calculated NO concentrations are almost constant. A maximum value of NO at $\phi = 1.0$ agrees well with the experiment, but about twice of the experimental value at $\phi = 1.2$. The peak position of NO at $\phi = 1.2$ is also faster than the experiment. No further attempt has been conducted for the improvement between simulation and experiment in the present study.

Ammonia combustion characteristics

Using the present model (UT-LCS), we investigated ammonia combustion characteristics like adiabatic flame temperature

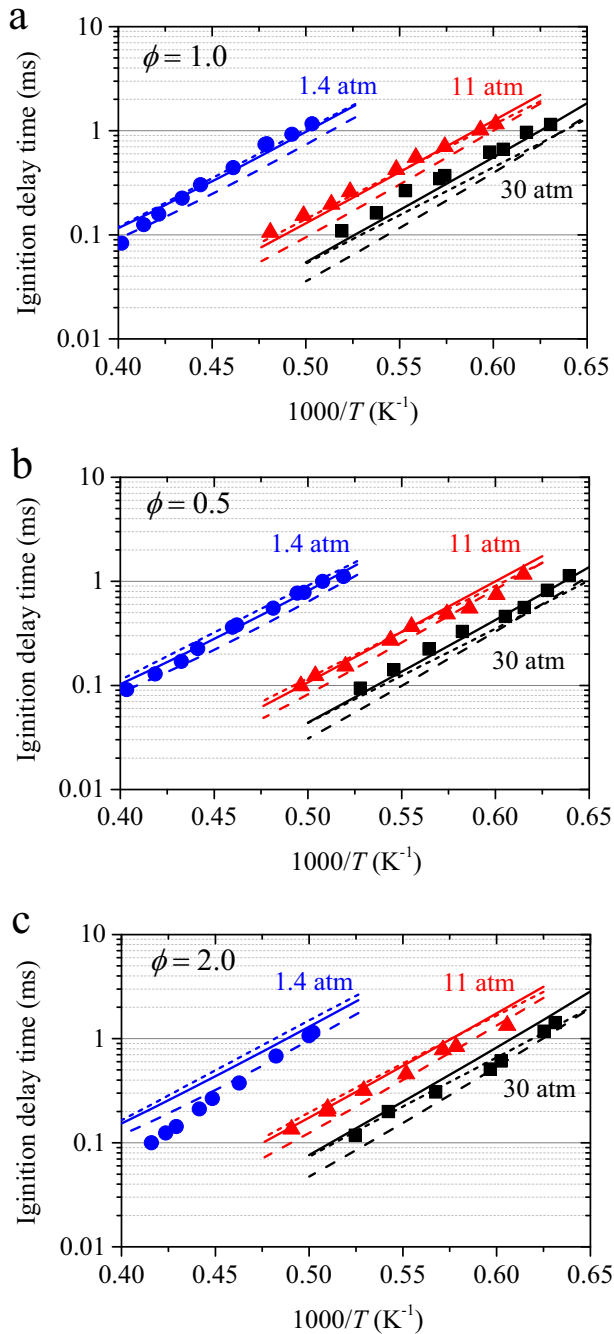


Fig. 1 – Ignition delay time with varieties of equivalence ratio ϕ (1.0–2.0), and pressure p (1.4–30 atm). (a) $\phi = 1.0$; (b) $\phi = 0.5$; (c) $\phi = 2.0$. Experimental values: ●, ▲, ■ (Mathieu and Petersen [18]). Calculated values: solid line: this work (UT-LCS model); broken line: the model of Song et al. [40]; dotted line: Mathieu and Petersen [18].

and reaction products. Fig. 3 shows the dependence of the adiabatic flame temperature in ammonia combustion T_b , on equivalence ratio ϕ , at atmospheric pressure. The maximum value of T_b was 2050 K at $\phi = 1$, which is lower than that of methane/air combustion by 100–150 K. Higher flame temperature may be desirable to the practical application of NH_3 combustion. In order to increase flame temperature, we

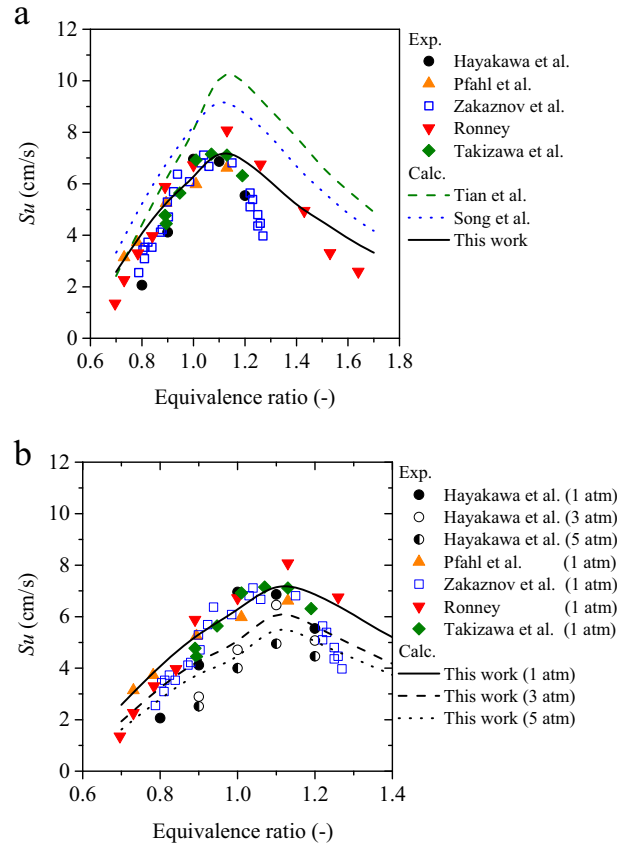


Fig. 2 – Laminar flame speed S_u vs. equivalence ratio at (a) atmospheric pressure and (b) high pressures. Comparison of experimental values with ammonia combustion models. Experimental values: Hayakawa et al. [19]; Pfahl et al. [48]; Zakaznov et al. [49]; Ronney [50]; Takizawa et al. [51]. Calculated values: this work (UT-LCS); Tian et al. [21]; Song et al. [40].

discuss ammonia/hydrogen combustion in the next section. The influence of equivalence ratio on the major species concentrations, NH_3 , H_2 , and NO was also depicted in Fig. 3. The concentration of NO decreased with increasing equivalence ratio, especially in the fuel rich region, $\phi > 1.3$, while the concentrations of hydrogen and residual ammonia increased in the fuel rich region, $\phi > 1.3$.

The detailed flame structures at $\phi = 0.7$, 1.1, 1.3, and 1.5 are shown in Fig. 4. Under a fuel-lean condition ($\phi = 0.7$), NO concentration downstream is very high. H_2 is produced at a flame front, but is burned at a post flame zone. At $\phi = 1.1$, H_2 is produced to very high concentration and it remains constant after the flame zone. NO concentration is still remained high (~500 ppm) downstream. At $\phi = 1.3$, NO concentration decreases to reach 30 ppm downstream, and H_2 is produced in large amount (~5%) by the decomposition of residual NH_3 . At $\phi = 1.5$, NO concentration decreases remarkably (~10 ppm) downstream, and H_2 is also produced in large amount (~5%). These features are suitable to practical applications. However, it is noted that a large part of residual NH_3 remains without decomposition. Under the fuel-rich condition ($\phi > 1.3$), relatively high concentration of NH_2 was observed downstream.

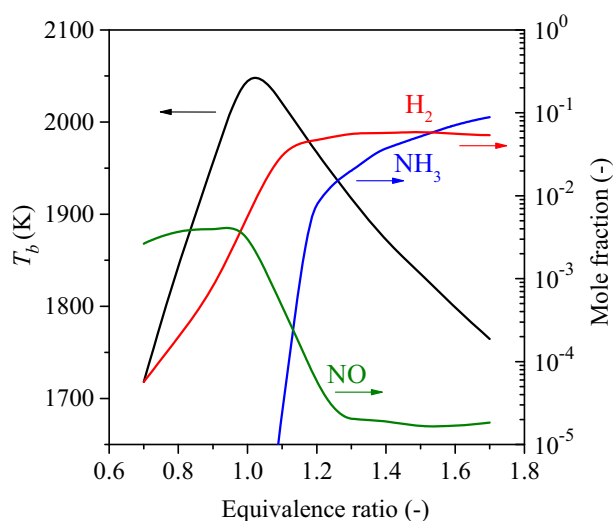


Fig. 3 – Adiabatic flame temperature as a function of equivalence ratio ϕ at 1 atm in NH_3 /air mixture, and mole fractions of major products vs. equivalence ratio downstream.

NH_2 plays a key role to reduce the concentration of NO, as discussed next.

The rate of production (ROP) analysis [63] for NO at $\phi = 0.7$ is shown in Fig. 5. NO is mainly formed via reactions of HNO and those of NH, but the Zeldovich mechanism (i.e., thermal NO; $\text{N} + \text{O}_2 \rightarrow \text{NO} + \text{O}$, $\text{O} + \text{N}_2 \rightarrow \text{NO} + \text{N}$, $\text{N} + \text{OH} \rightarrow \text{NO} + \text{H}$) hardly contributes to NO formation. This is because flame temperature is relatively low (~ 1720 K in Fig. 3a). NO, which is

formed from HNO or NH, is consumed by the reactions with NH_2 and NH radicals (i.e., DeNOx process) early in the reaction zone. On the other hand, NO formation rate via reactions of HNO is higher than NO consumption rate described above at the latter part in the reaction zone, resulting in the total NO formation rate becoming positive and NO concentration increasing downstream.

Under a fuel-rich condition ($\phi = 1.3$), total NO formation rate becomes negative at a latter part in the reaction zone and NO concentration decreases downstream. NO formation via $\text{NO}_2 + \text{H}$ contributes to increasing NO concentration as well as NO formation via $\text{HNO} + \text{H}$ and $\text{HNO} + \text{O}_2$. The Zeldovich mechanism ($\text{N} + \text{O}_2 \rightarrow \text{NO} + \text{O}$) also contributes to NO formation at a latter part in the reaction zone and the inverse Zeldovich mechanism can proceed ($\text{N} + \text{NO} \rightarrow \text{N}_2 + \text{O}$). NO decreases via reaction with NH_2 , and it decreases further via the reaction of $\text{NH} + \text{NO} \rightarrow \text{N}_2 + \text{OH}$, resulting in the decrease of NO concentration downstream.

In NH_3 /air combustion, NH_3 -lean conditions ($\phi < 1$) are impractical because NO concentration reaches several percent in the exhaust gas. Under NH_3 -rich conditions ($\phi > 1.3$), NH_3 is not fully decomposed and thus several percent of NH_3 is included in the exhaust gas although NO concentration in the exhaust gas is very low. The flame speed is also slow in NH_3 /air combustion. To improve these combustion characteristics, we investigated NH_3/H_2 /air mixture, as described in the next section.

Ammonia/hydrogen combustion characteristics

To improve the characteristics of ammonia combustion, we investigated ammonia/hydrogen combustion. First, we

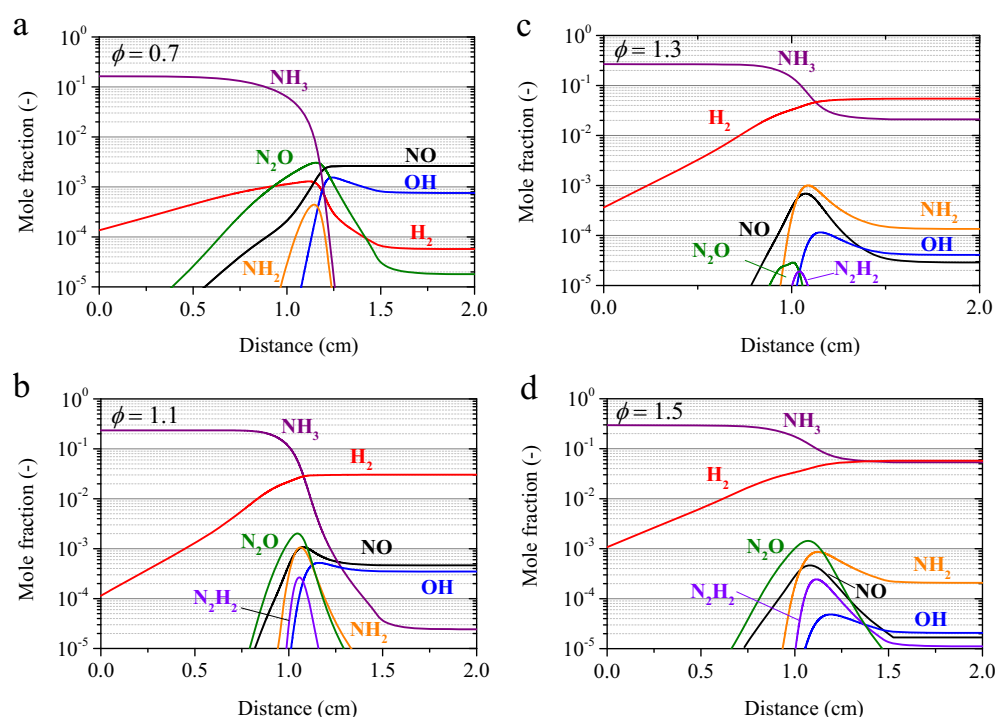


Fig. 4 – Variations of mole fractions in flame structures with a variety of equivalence ratios ($\phi = 0.7, 1.1, 1.3$, and 1.5) in NH_3 /air at 1 atm.

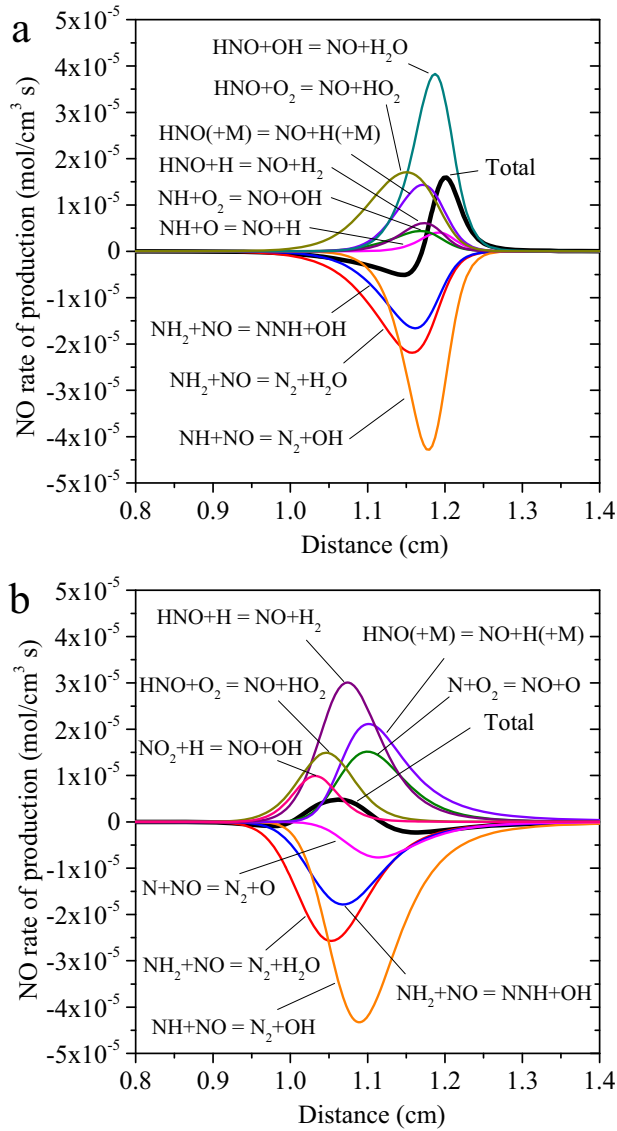


Fig. 5 – Rate of production (ROP) analysis for NO in NH_3/air at (a) $\phi = 0.7$ and (b) $\phi = 1.3$.

checked the validity of the present model by comparing with experimental data of laminar flame velocity in $\text{NH}_3/\text{H}_2/\text{air}$ mixtures under atmospheric and pressurized conditions from 1 to 5 atm. The present model agrees slightly better with experimental data over at $p = 3$ and 5 atm in comparison with previous models (Fig. 6), while the previous models by Tian et al. [21] and Song et al. [40] predicted the experimental data at 1 atm more correctly than the present model. Thus, the present model predicted the experimental data well at high pressures. In low mole fraction regions of H_2 (<0.2), the present model demonstrated the experimental data at 3 and 5 atm more correctly than previous models, such as that of Tian et al. [21], because the reaction mechanism of NH_3 combustion was modified in the present model as discussed in the previous section. In the region of high H_2 mole fraction, however, the difference between the present model and the previous model was small, because the $\text{NH}_3/\text{H}_2/\text{air}$ mixture is governed by H_2 combustion.

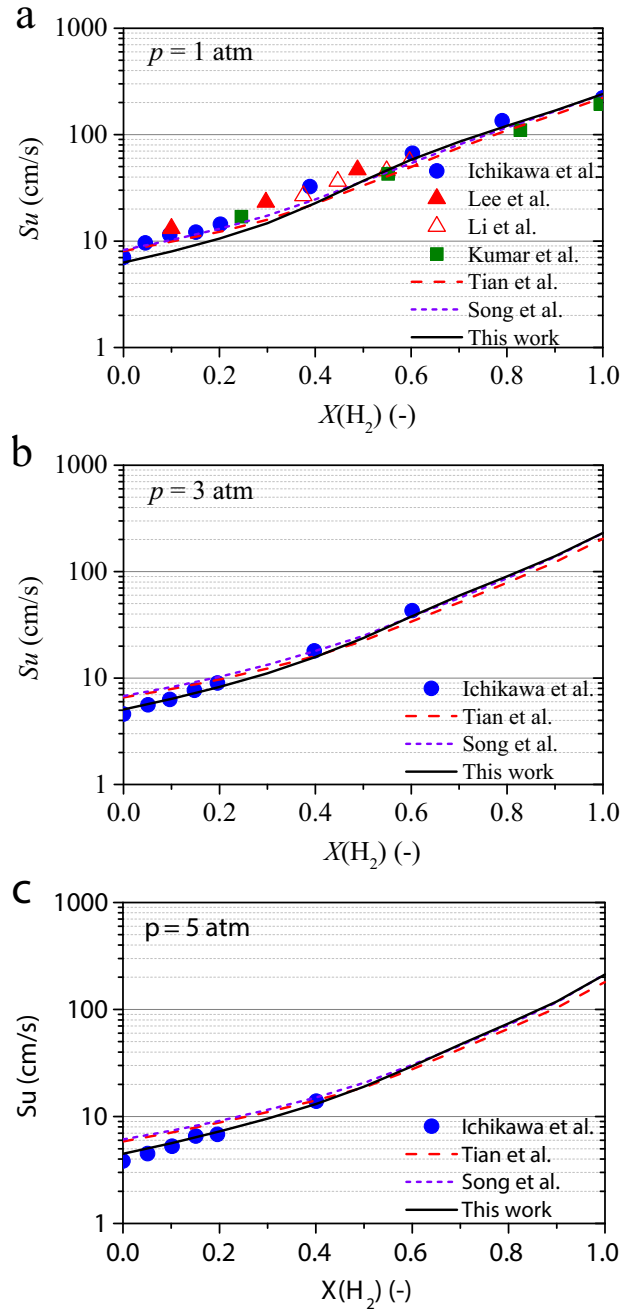


Fig. 6 – Laminar flame velocity in $\text{NH}_3/\text{H}_2/\text{air}$ mixtures ($\phi = 1$) under atmospheric and pressurized conditions. (a) 1 atm; (b) 3 atm; (c) 5 atm. $X(\text{H}_2) = [\text{H}_2]_0 / ([\text{NH}_3]_0 + [\text{H}_2]_0)$. Experimental values: Ichikawa et al. [29], Lee et al. [32], Li et al. [30], and Kumar et al. [26]. Calculated values: solid line: this work (UT-LCS); broken line: Tian et al. [21]; dotted line: Song et al. [40].

The effects of H_2 addition on laminar flame velocity, adiabatic flame temperature, and NO mole fraction were depicted as a function of equivalence ratio in Fig. 7, in which NO concentrations were taken at $x = 2$ cm where the NO concentrations are in steady state (see Fig. 7c), as shown in

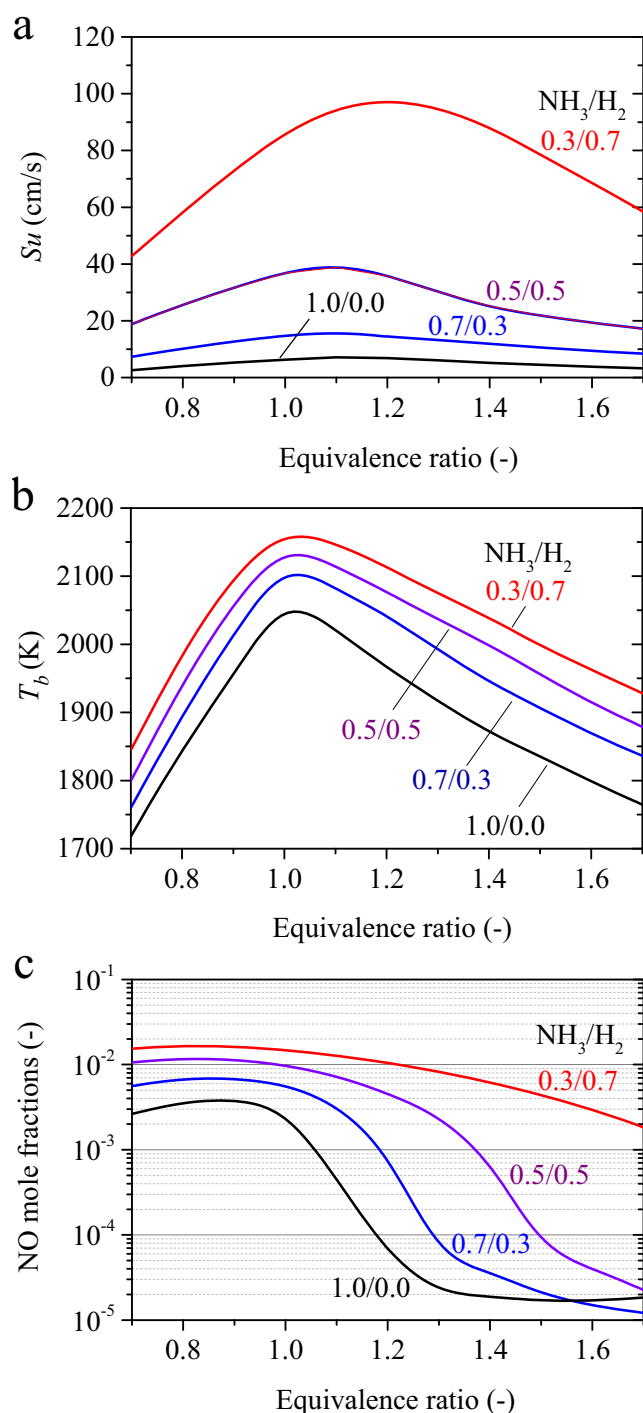


Fig. 7 – Effects of H₂ addition on (a) laminar flame velocity, S_u , (b) adiabatic flame temperature, T_b , and (c) NO mole fraction downstream were calculated as a function of equivalence ratio (Total pressure: 1 atm) in NH₃/H₂/air mixtures.

Fig. 4. The laminar velocity and the adiabatic flame temperature increased with increasing H₂ mole fraction. NO concentration increased with an increase in H₂ mole fraction, mainly owing to thermal NOx. In addition, we found that the fuel rich condition (i.e., equivalence ratio, $\phi > 1.3$) is

suitable to suppress NO formation in the order of several tens of ppm.

We also evaluated the ratios of NH₃ and H₂ concentrations downstream to initial NH₃ and H₂ concentrations, i.e., $[\text{NH}_3]/[\text{NH}_3]_0$ and $[\text{H}_2]/[\text{H}_2]_0$, as a function of equivalence ratio (Fig. 8). NH₃ and H₂ concentrations downstream was taken at $x = 2$ cm where NH₃ and H₂ concentrations are in steady state, as shown in Fig. 4. With increasing H₂ in the NH₃/H₂/air mixture, $[\text{NH}_3]/[\text{NH}_3]_0$ was decreased and $[\text{H}_2]/[\text{H}_2]_0$ was increased in $\phi > 1.3$. As can be seen in Fig. 7c, however, thermal NO increased with increasing H₂ addition. Considering the requirements of low NO formation and low residual NH₃ in exhaust gas, an equivalence ratio ϕ of around 1.3 is suitable. Therefore, treatment of H₂ and NH₃, which are found in relatively high concentrations in exhaust gas (e.g., NH₃:H₂ = 0.7/0.3 with ϕ of around 1.3), should be considered in new combustion systems. A fuel-rich condition like $\phi \approx 1.3$ is effective to reduce NOx and available for new combustors such as an exhaust gas recirculation (EGR) system and a staged combustion system, in which high power and low NOx emission are expected because of relatively high H₂ concentration and low NO concentration [64].

Fig. 9a shows the flame structure at $\phi = 1.3$ with NH₃/H₂ = 7/3 in an NH₃/H₂/air mixture. NO concentration decreased downstream adequately as well as in NH₃/air. In addition, we found that H₂ concentration decreased at around 1 cm of distance and recovered downstream below 1 cm of distance (see the details in Fig. 9b). This suggests that NH₃ decomposition was induced in the reaction zone.

Fig. 10 shows a ROP analysis for NO formation in an NH₃/H₂/air mixture. NO formation is due to the reaction via HNO as well as NH₃ combustion in NH₃/air. However, because the flame temperature increases in the NH₃/H₂/air mixture, NO formation due to thermal NOx (i.e., $\text{N} + \text{O}_2 \rightarrow \text{NO} + \text{O}$, $\text{N} + \text{OH} \rightarrow \text{NO} + \text{H}$) contributes to the increase of NO concentration. Because NO concentration is decreased via $\text{N} + \text{NO} \rightarrow \text{N}_2 + \text{O}$, as well as the reactions with NH₂, NO concentration is totally decreased downstream.

H₂ formation in the NH₃/H₂/air mixture was also analyzed by ROP (Fig. 11). The ROP analysis suggests the mechanism of H₂ consumption and regeneration. The following reactions of H₂ oxidation and chain-branching contribute for H₂ consumption, ($\text{O} + \text{H}_2 \rightarrow \text{OH} + \text{H}$, $\text{OH} + \text{H}_2 \rightarrow \text{H} + \text{H}_2\text{O}$). Meanwhile, the reactions between NH_x and H contribute to H₂ regeneration ($\text{NH}_3 + \text{H} \rightarrow \text{NH}_2 + \text{H}_2$, $\text{NH}_2 + \text{H} \rightarrow \text{NH} + \text{H}_2$, $\text{NH} + \text{H} \rightarrow \text{N} + \text{H}_2$, and $\text{N} + \text{N} + \text{M} \rightarrow \text{N}_2 + \text{M}$). The total reaction is: $\text{NH}_3 + 3\text{H} \rightarrow 1/2\text{N}_2 + 3\text{H}_2$. Thus NH₃ decomposition is induced by these mechanism. In addition, HNO and N₂H₂ radicals can contribute to H₂ formation ($\text{HNO} + \text{H} \rightarrow \text{NO} + \text{H}_2$, $\text{N}_2\text{H}_2 + \text{H} \rightarrow \text{NNH} + \text{H}_2$).

As discussed above, the addition of H₂ in NH₃ fuels can increase flame temperature as well as increasing flame speed. Also, under fuel-rich conditions, adequately low concentrations of NO and NH₃ and relatively high concentration of H₂ can be obtained in exhaust gas by controlling an appropriate ratio of NH₃/H₂ (e.g., $\phi \approx 1.3$ with NH₃/H₂ = 0.7/0.3–0.5/0.5). In fact, high NO_x formation was reported under a fuel-lean condition using a swirling gas turbine combustor with a NH₃/H₂/air mixture [65]. Therefore, combustion under the fuel-rich condition with a NH₃/H₂/air mixture will be effective in reducing NO concentration as well as H₂ formation from

NH_3 , H_2 formation will assist to improve performance in combustors. This feature can be applied to new combustion systems such as EGR and staged combustion systems, as mentioned above. To use this feature of $\text{NH}_3/\text{H}_2/\text{air}$

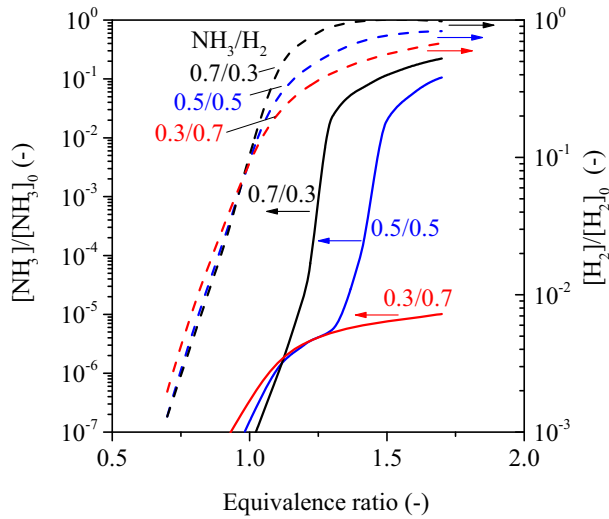


Fig. 8 – Ratios of $[\text{NH}_3]/[\text{NH}_3]_0$ and $[\text{H}_2]/[\text{H}_2]_0$ vs. function of equivalence ratio.

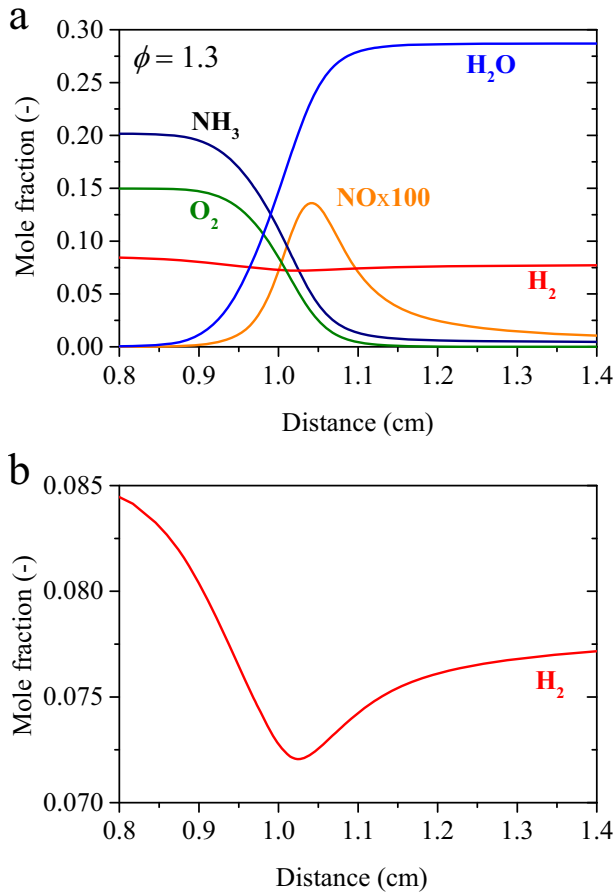


Fig. 9 – Flame structure for varieties of mole fractions of (a) NH_3 , H_2 , O_2 , H_2O , NO , OH and (b) enlarged graph of H_2 at $\phi = 1.3$ with $\text{NH}_3/\text{H}_2 = 0.7/0.3$.

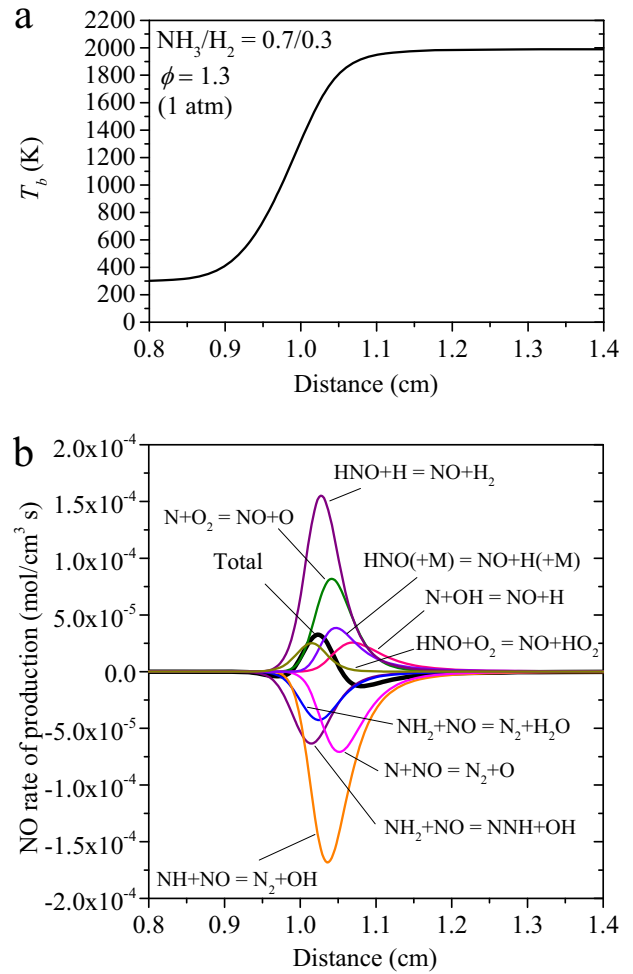


Fig. 10 – Temperature profile and ROP analysis for NO in an $\text{NH}_3/\text{H}_2/\text{air}$ mixture at $\phi = 1.3$.

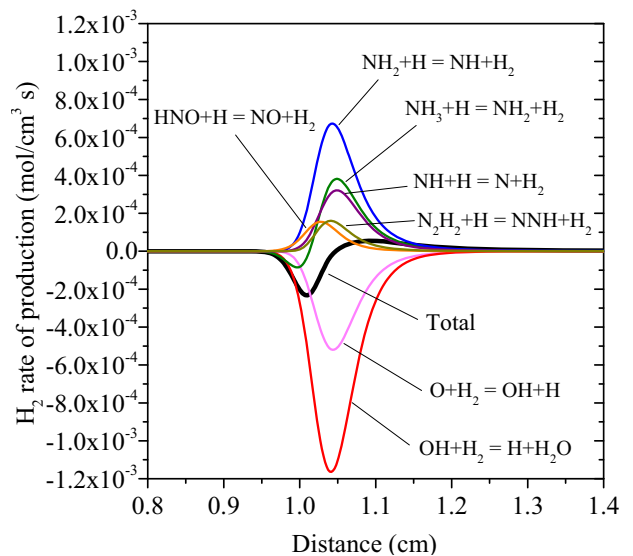


Fig. 11 – Temperature profile and ROP analysis for H_2 in an $\text{NH}_3/\text{H}_2/\text{air}$ mixture at $\phi = 1.3$.

combustion effectively, new combustion systems should be developed for future applications.

Conclusion

The reaction mechanism of ammonia oxidation based on a previous literature (i.e., the model of Song et al. [40]) was improved to predict the characteristics of ammonia combustion. In the present model (UT-LCS), elementary reactions including NH_2 , HNO , and N_2H_2 play key roles in improving the model. The present UT-LCS model reproduced the experimental data of laminar flame speed and ignition delay time in wide ranges of equivalence ratio and pressure very well. Using this model, NH_3/H_2 combustion was also investigated. The model explains experimental data of laminar flame speed in a wide range of equivalence ratios correctly. The model suggested appropriate conditions to reduce NO concentration in NH_3/H_2 combustion. We also found H_2 formation via NH_3 decomposition occurs in NH_3/H_2 mixed fuel using this model. These results can contribute to effective combustion of NH_3/H_2 air with a new combustor for future applications.

Appendix A. Supplementary data

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.ijhydene.2017.12.066>.

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